1 **(a)** 5.2 **[B1]**

(b) 4700 **[B1]**

(c) 850 **[B1]**

[Total Marks: 3]

2 **(a)** $80 \times 45 = 3600$ **[M1]**

0.36 **[A1]**

(b) 3200 **[B1]**

[Total Marks: 3]

3 **(a)** 35000 **[B1]**

(b) $60 \times 45 \times 30$ OR 2700×30 OR 81000 **[M1]**

their $81000 \div 1000$ **[M1]**

81 **[A1]**

[Total Marks: 4]

4 **(a)** 4750 **[B1]**

(b) 0.32 **[B1]**

(c) 24 000 **[B1]**

(d) 6.5 **[B1]**

[Total Marks: 4]

5 **(a)** Converts units correctly to cm: 250 mm = 25 cm AND 0.2 m = 20 cm **[B1]**

20000 **[A1]**

(b) 20 **[B1]**

(c) Uses Mass = Density × Volume: 0.6×20000 (= 12000 g) **[M1]**

12 **[A1]**

[Total Marks: 5]

6 **(a)** **(i)** $2(45 \times 30 + 45 \times 20 + 30 \times 20) = 2(1350 + 900 + 600)$ **[M1]**

5700 **[A1]**

(ii) 0.57 **[B1]**

(b) Volume $= 45 \times 30 \times 20 = 27\,000\ cm^3$ **[B1]**

Capacity = 27 litres **[B1]**

(c) 650 g = 0.65 kg and $18 \times 1.15 = 20.7$ kg **[M1]**

21.35 **[A1]**

[Total Marks: 7]

7 **(a)** Perimeter on plan $= 2(12 + 8) = 40$ cm and actual perimeter $= 40 \times 25\,000 = 1\,000\,000$ cm **[M1]**

10 **[A1]**

(b) **(i)** $(12 \times 8) - (4 \times 1.5) = 96 - 6$ **[M1]**

90 **[A1]**

(ii) 1 cm represents 0.25 km, so 1 cm^2 represents $0.25^2 = 0.0625$ km^2 (or $90 \times 25\,000^2 = 5.625 \times 10^{10}$ cm^2) **[M1]**

5.625 **[A1]**

(c) Actual pond dimensions: $4 \times 250 = 1000$ m and $1.5 \times 250 = 375$ m **[M1]**

Volume $= 1000 \times 375 \times 1.6 = 600\,000$ m^3 **[M1]**

6×10^8 **[A1]**

[Total Marks: 9]

8 **(a)** Converts all dimensions to cm: 250, 140, 0.8 **[M1]**

$250 \times 140 \times 0.8 = 28000$ **[A1]**

(b) Multiplies volume by 7.8: $28000 \times 7.8 = 218400$ **[M1]**

218.4 **[A1]**

[Total Marks: 4]

9 **(a)** Converts dimensions consistently, e.g. 1.8 m = 180 cm (or 0.75 m and 0.6 m) **[B1]**

Calculates volume: $180 \times 75 \times 60 = 810\,000$ cm^3 (or 0.81 m^3) **[M1]**

810 **[A1]**

(b) Converts 810 litres = 810 000 ml (or 450 ml/s = 0.45 l/s) **[M1]**

Calculates time in seconds: $\dfrac{810\,000}{450} = 1800$ seconds **[M1]**

30 **[A1]**

[Total Marks: 6]

10 **(a)** Converts dimensions to consistent units, e.g. 150 cm, 80 cm, 60 cm OR 1.5 m, 0.8 m, 0.6 m **[M1]**

$150 \times 80 \times 60 = 720000$ OR $1.5 \times 0.8 \times 0.6 = 0.72$ **[M1]**

720 **[A1]**

(b) $0.024 \times 1000000 = 24000$ **[M1]**

Mass $= 24000 \times 7.8 = 187200$ **[M1]**

187.2 **[A1]**

[Total Marks: 6]

11 (a) Conversion of units: 15 $mm = 0.015$ m OR 15 $m = 1500$ cm etc. **[M1]**

Volume $= 15 \times 12 \times 0.015$ (or eq) **[M1]**

2700 **[A1]**

(b) Use of consistent units for tank volume equation (e.g. $r = 0.6$ m and $V = 2.7$ m^3 OR $V = 2700000$ cm^3 and $r = 60$ cm) **[M1]**

$\pi \times 0.6^2 \times h = 2.7$ OR $\pi \times 60^2 \times h = 2700000$ **[M1]**

$h = \dfrac{2.7}{0.36\pi}$ (m) OR $h = \dfrac{2700000}{3600\pi}$ (cm) **[M1]**

238.7 **[A1]**

[Total Marks: 7]

12 (a) $\dfrac{4}{3} \times \pi \times 15^3$ OR 4500π OR 14137 **[M1]**

their $14137 \div 1000000$ OR $\dfrac{4}{3} \times \pi \times 0.15^3$ **[M1]**

1.41×10^{-2} **[A1]**

(b) 125 kg = 125000 g **[B1]**

$\dfrac{125000}{4500\pi}$ OR $\dfrac{125000}{14137}$ **[M1]**

8.84 (8.841 . . .) **[A1]**

(c) (i) 24 mm = 2.4 cm **[B1]**

$\dfrac{1}{3} \times \pi \times 2.4^2 \times 5.5$ **[M1]**

33.2 (33.17 . . .) OR 10.56π **[A1]**

(ii) $\dfrac{4500\pi}{\text{their } 33.17}$ OR 426.1 . . . **[M1]**

426 **[A1]**

[Total Marks: 11]

13 (a) Identifies consistent units for radius and length (e.g. $r = 1.5$ cm and length = 80 cm) **[M1]**

Volume in one second $= \pi \times 1.5^2 \times 80 = 180\pi$ cm^3 $(= 565.48 \ldots$ $cm^3)$ **[M1]**

Converts to litres $(180\pi \div 1000)$ and confirms $0.56548 \cdots \approx 0.565$ **[A1]**

(b) Volume of pool required $= 4 \times 2.5 \times 1.2 = 12$ m^3 OR 12000 litres **[M1]**

Time in seconds $= 12000 \div 0.56548 \ldots$ OR $12000 \div \dfrac{180\pi}{1000}$ **[M1]**

$= 21220.6\ldots$ seconds ($353.6\ldots$ minutes) **[A1]**

5 hours 54 minutes **[A1]**

[Total Marks: 7]

14 **(a)** Converts speed to cm/min: $5.4 \times 1000 \times 100 \div 60 = 9000$ **[M1]**

Calculates volume in 1 minute: $\pi \times 2.5^2 \times 9000 = 56250\pi\ cm^3$ **[M1]**

Converts cm^3 to litres and evaluates: $56250\pi \div 1000 = 176.714\ldots$ **[A1]**

(b) Calculates total volume in 45 minutes: $56250\pi \times 45 = 2531250\pi\ cm^3$ (or uses litres: $176.714\cdots \times 45 = 7952.1\ldots$ litres) **[M1]**

Converts tank base dimensions to cm and finds area: $300 \times 200 = 60000\ cm^2$ **[M1]**

Sets up equation for depth: Depth $= \dfrac{2531250\pi}{60000}$ **[M1]**

133 (or 132.5 to 132.6) **[A1]**

[Total Marks: 7]

15 **(a)** Calculates area in m^2: $450 \times 320 = 144\,000$ m^2 (or 0.45×0.32) **[M1]**

0.144 **[A1]**

(b) **(i)** Total volume in ml $= 144\,000 \times 25 = 3\,600\,000$ ml **[M1]**

Converts ml to litres or cm^3: 3600 litres or $3\,600\,000$ cm^3 **[M1]**

3.6 **[A1]**

(ii) Mass of liquid in one drum $= 216 - 9 = 207$ kg $= 207\,000$ g **[B1]**

Volume of one drum $= 0.18$ $m^3 = 180\,000$ cm^3 **[B1]**

$\dfrac{207\,000}{180\,000} = 1.15$ **[A1]**

[Total Marks: 8]

16 **(a)** Volume $= 15.7\ m^3$ **[B1]**

$\pi r^2(2.5r) = 15.7$ giving $r^3 = \dfrac{15.7}{2.5\pi}$ $(= 1.99898\ldots)$ **[M1]**

$r = 1.26$ **[A1]**

(b) Internal surface area $= 2\pi(1.26)^2 + 2\pi(1.26)(3.15) = 34.913\ldots\ m^2$ $(= 349\,131\ldots\ cm^2)$ **[M1]**

Volume $= 349\,131\cdots \times 0.15$ (or $34.913\cdots \times 0.0015 \times 10^6$) **[M1]**

$52\,400$ **[A1]**

(c) $15\,700$ litres $= 15\,700\,000$ ml and time $= \dfrac{15\,700\,000}{450} = 34\,888.88\ldots$ seconds **[M1]**

$\dfrac{34\,888.88\ldots}{60} = 581.48\ldots$ minutes **[M1]**

9 hours 41 minutes **[A1]**

(d) Mass $= 15\,700\,000 \times 1.02 = 16\,014\,000$ g $(= 16\,014$ kg) **[M1]**

16.014 **[A1]**

[Total Marks: 11]

17 (a) Cross-sectional area $= \pi \times 2.5^2$ $(= 6.25\pi)$ **[M1]**

Speed conversion: $18\ km/h = \dfrac{18 \times 1000 \times 100}{60}\ cm/min$ $(= 30000\ cm/min)$ **[M1]**

Volume per minute $= 6.25\pi \times 30000$ $(= 187500\pi\ cm^3)$ **[M1]**

Convert to litres: $\dfrac{187500\pi}{1000}$ **[M1]**

589 or 589.0 to 589.1 (187.5π) **[A1]**

(b) Capacity of reservoir $= 50 \times 30 \times 4 = 6000\ m^3$ OR 6000000 litres **[M1]**

Time in minutes $= \dfrac{6000000}{187.5\pi}$ (or 6000000÷ their (a)) $(= 10185.9...)$ **[M1]**

Time in hours $= \dfrac{10185.9...}{60}$ $(= 169.76...)$ **[M1]**

$169.76... \div 24 = 7.07...$ days so 7 full days **[M1]**

$0.07... \times 24 = 1.76...$ hours (or $169.76... - 7 \times 24$) **[M1]**

7 **[A1]**

2 **[A1]**

[Total Marks: 12]

18 (a) $\pi \times 2.5^2 \times 160$ OR $\pi \times 0.025^2 \times 1.6$ **[M1]**

$\times 60$ **[M1]**

$\div 1000$ (if using cm) OR $\times 1000$ (if using m) **[M1]**

188 $(188.4\ldots)$ **[A1]**

(b) Volume of tank $= 4.2 \times 3.5 \times 2.4 = 35.28\ m^3$ **[M1]**

$35.28 \times 1000 = 35280$ litres **[M1]**

Time $= \dfrac{35280}{\text{their } 188.49}$ OR $\dfrac{35280}{60\pi}$ **[M1]**

187 minutes $(187.1\ldots)$ **[A1]**

3 hours 7 minutes **[A1]**

(c) Calculates lateral area correctly: $2 \times 4.2 \times 2.4 + 2 \times 3.5 \times 2.4$ OR $36.96\ m^2$ [M2]
(If incorrect, M1 for one correct face area e.g., 4.2×2.4)

$36.96 \times 10000 = 369600\ cm^2$ OR $45000 \div 10000 = 4.5\ m^2$ **[M1]**

Litres of paint required $= \dfrac{369600}{45000}$ OR $\dfrac{36.96}{4.5}$ **[M1]**

Number of tins $= \dfrac{8.21\ldots}{2.5}$ $(= 3.28\ldots)$ **[M1]**

4 **[A1]**

[Total Marks: 15]